WELLESLEY

Gender Differences in Time Spent on Education

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Introduction/Background

- Studies and news articles have highlighted how boys are falling behind girls in school with
  - lower reading scores (New York Times)
  - a graduation rate of 83% for boys and 89% for girls in 2023 (Brookings)
  - 66% women enrolled in college compared to 57% of men in 2022 (National Center for Education Statistics).

To investigate if this trend manifests itself in how much time students spend on education, I seek to answer the following questions:

Has time spent on educational activities varied by gender from 2003-2024? Are there other factors, including income, location, race or their interactions with gender that contribute to differences in time spent on education?

Data

Data Sources

- Roster, Respondent, Activity, and Current Population Survey files from the American Time Use Survey Data (ATUS) spanning 2003-2024.

Sample

- 16,898 full-time students aged 15-24 in the ATUS dataset

Data Cleaning

- To define educational activities, I included all categories under the major category of education with the label 06. The main activities include taking class, extracurricular club activities excluding sports, and homework.
- I summed all the time people spent on educational activities in a day, and filled in 0's for people who had no recorded time spent on education.
- I then aggregated several categorical variables into broader categories to avoid having too many levels, including income to 5 levels, race to 6 levels, and year into 4 periods of 4 years each with the last period being 6 years.

Models

Predictors

- Primary Predictor: Gender
- Secondary Predictors: Household Family Income, Race, Region
- Interactions w/ gender: Race, Income, Region, Year, College vs. High School
- Other predictors to control for: Weekday vs. Weekend, Month, Period, College vs. High School

- **4 models: 2 logistic regression models and 2 linear regression models**
- Each model type is run twice with:

1. A focus on gender differences with the variable period (aggregated years).
2. An interaction between numeric year and gender to see if there is change over time by gender.

- **Logistic Regression Model Outcome:** Whether or not the student spent any time on educational activities at all
- **Linear Regression Model Outcome:** For students who spent any time on education (n = 9024), the logged minutes spent on educational activities

Visualizations

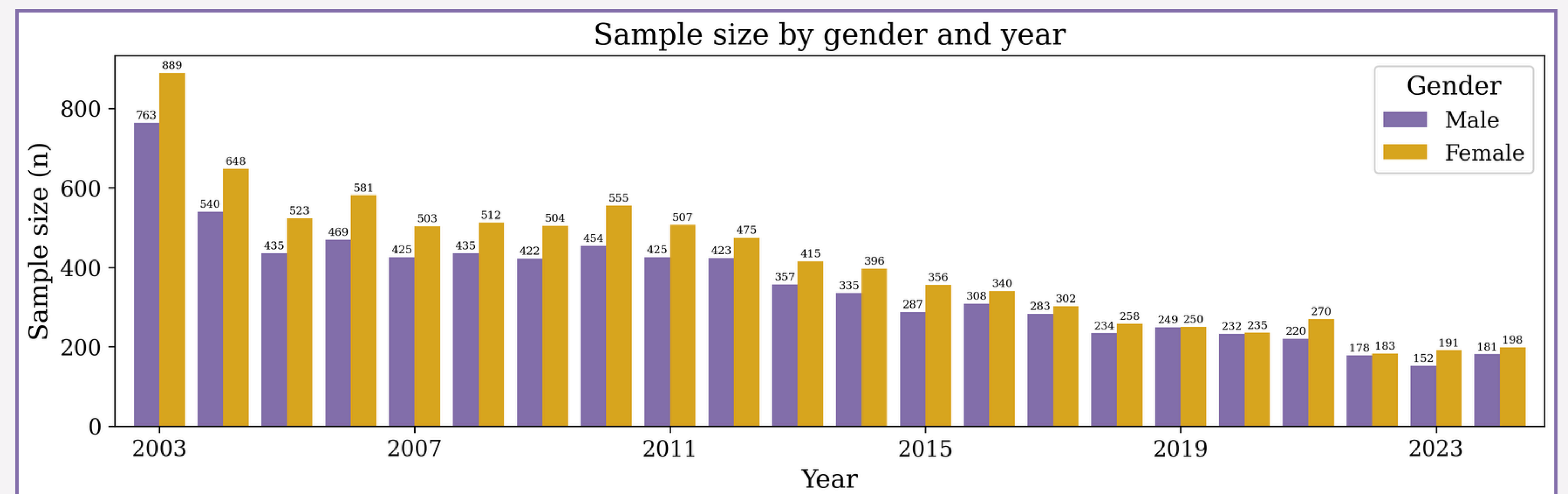

Figure 1: Sample size decreasing over time.

Results

ANOVA P-Values	Model 1: Logistic Regression with Periods	Model 2: Logistic Regression with Numeric Year	Model 3: Logistic Regression with Periods	Model 4: Logistic Regression with Numeric Year
Gender	Uninterpretable	0.677	Uninterpretable	0.602
Year		0.196		0.542
Period (aggregated years)	< 0.0001		0.846	
Race	Uninterpretable	Uninterpretable	Uninterpretable	Uninterpretable
Family Income	Uninterpretable	Uninterpretable	Uninterpretable	Uninterpretable
Region	Uninterpretable	Uninterpretable	Uninterpretable	Uninterpretable
College v. High School	Uninterpretable	Uninterpretable	Uninterpretable	Uninterpretable
Weekend/ Weekday	< 0.0001	< 0.0001	< 0.0001	< 0.0001
Month	< 0.0001	< 0.0001	< 0.0001	< 0.0001
Gender x Year		0.699		0.612
Gender x Race	0.843	0.839	0.336	
Gender x Income	0.446	0.512	0.191	0.227
Gender x Region	0.351	0.353	0.53	0.325
Gender x School Level	0.016	0.021	0.032	0.028

ANOVA P-Values	Model 1: Logistic Regression with Periods	Model 2: Logistic Regression with Numeric Year	Model 3: Logistic Regression with Periods	Model 4: Logistic Regression with Numeric Year
Gender + interactions with Gender	< 0.0001	< 0.0001	0.027	0.038
Year (numeric) + interactions		< 0.0001		0.662
Race + interactions	< 0.0001	< 0.0001	< 0.0001	< 0.0001
Race + interactions	< 0.0001	< 0.0001	0.02	0.023
Region + interactions	0.008	0.007	0.53	0.53
College v. High School + interactions	< 0.0001	< 0.0001	< 0.0001	< 0.0001

Logistic Regression- Variables of Interest	Odds Ratio	P-Value	95% Confidence Interval
Female	1.435	0.003	(1.127, 1.828)
Asian	1.722	< 0.0001	(1.394, 2.133)
Female x College	0.845	0.016	(0.74, 0.97)
2008-2012	1.12	0.009	(1.028, 1.219)
2008-2012	1.215	< 0.0001	(1.105, 1.337)
2008-2012	1.165	0.003	(1.054, 1.288)

Linear Regression	Coefficient Estimate	P-Value	95% Confidence Interval
Female	-8.04	0.145	(-17.66, 2.71)
Asian	20.78	< 0.0001	(10.69, 31.79)
Weekend	-53.98	< 0.0001	(-55.65, -52.24)
Female x Hispanic	9.83	0.04	(0.29, 20.29)

Multiple R-squared: 0.21

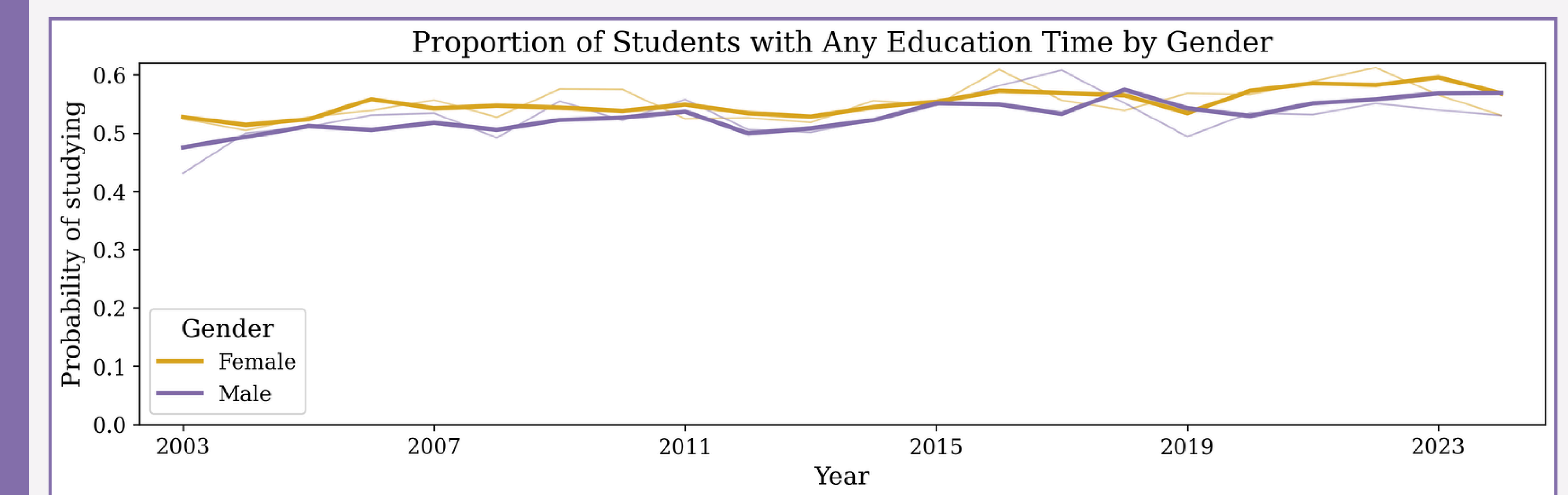

Figure 2: Probability of studying at all is statistically significantly higher for women.

Note: Lighter lines = actual proportions and darker lines = predicted probabilities

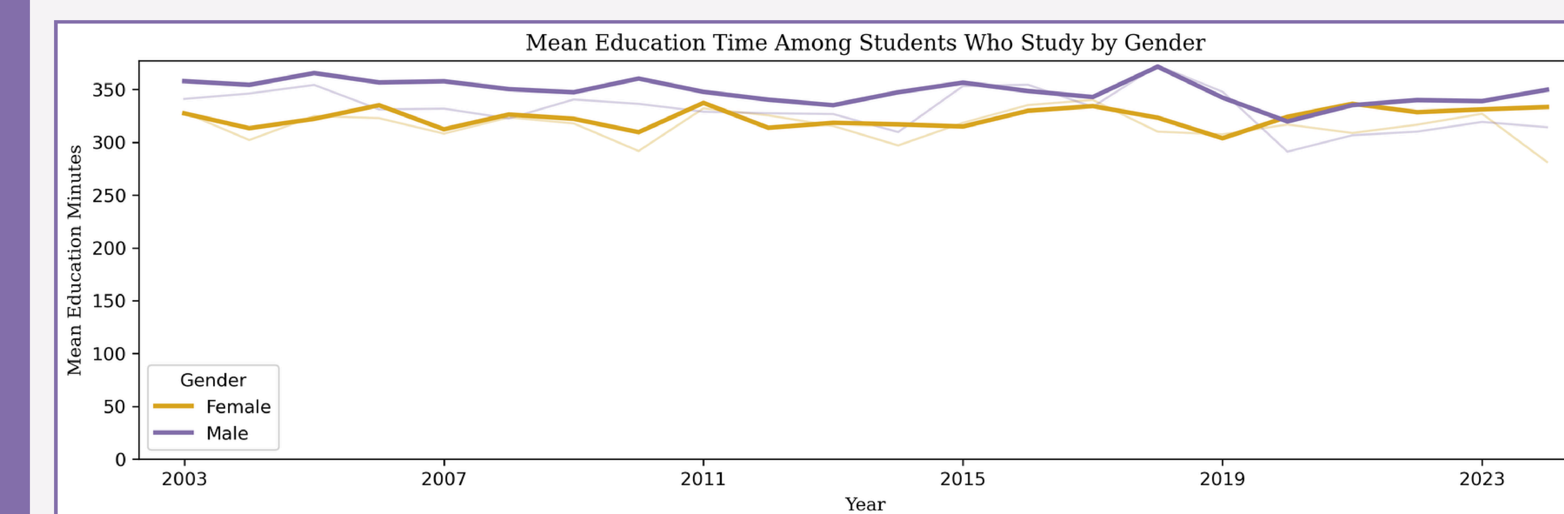

Figure 3: No statistically significant difference between men and women in amount of time studied if they do study.

Note: Lighter lines = actual means and darker lines = predicted means

Discussion and Conclusion

- There is **a statistically significant difference between boys and girls in whether or not they spend time on education at all**. Girls have a 43.5% higher overall odds of studying on their survey day compared to boys.
- However, there is no statistically significant gender gap in how much time students spend on educational activities if they did engage in them.

Limitations

- ATUS sample size is decreasing across years, which possibly affects the year variation change in education time.
- Near equal amount of weekends and weekdays in the ATUS dataset- weekend overrepresentation and less time spent on educational activities.
- Family Income is grouped into fixed buckets
- Mild heteroskedasticity for the linear regression model, as there is a lot of variability in educational time across levels of independent variables
  - Handled by HC3, a type of heteroskedasticity consistent robust standard errors, which manage high leverage observations and is used for small- medium sample sizes

Future work

- Disaggregate Asian by specific race
- Explore other interactions, such as interactions for race and income

References

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